MPSA 2026 - estimation results for paper

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1. Set up

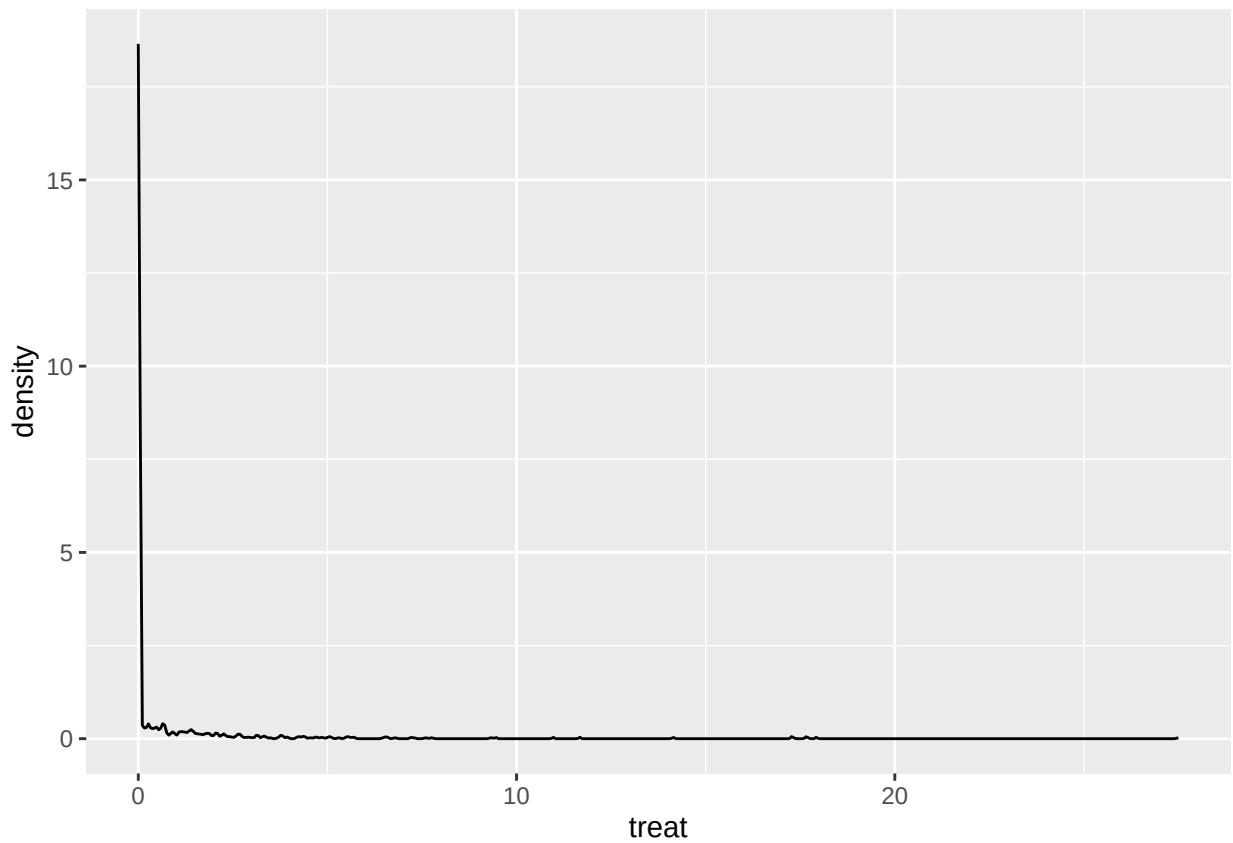
2. Data cleaning

Treatment is disappearances and homicides by the state; calculated as the yearly rate by 10,000

2.1 choosing treatment and covariates

2.2 Covariates

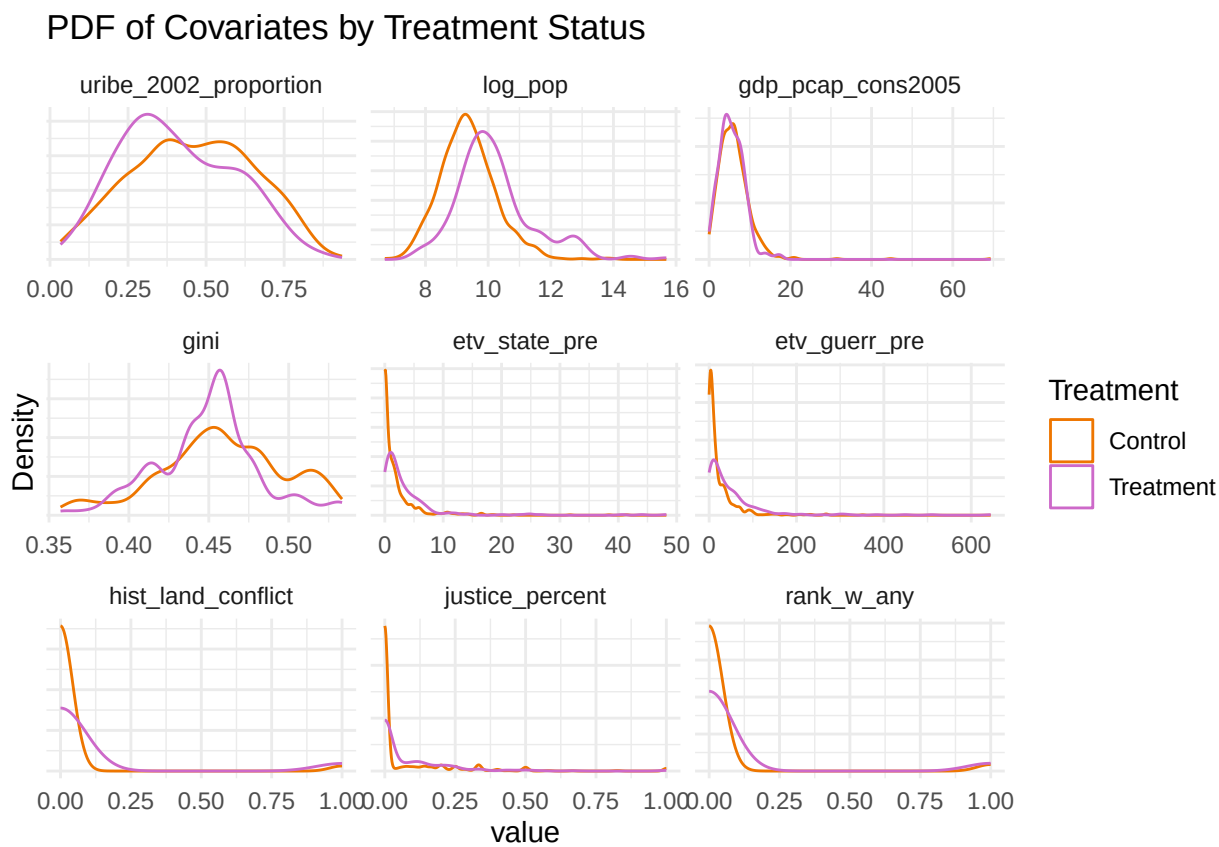
A density plot of the yearly rate of cases per 10,000 per municipality; followed by the exact number and proportion of control municipalities




```
##    0
## 746
```

```
## [1] 0.7415507
```

2.2.1 Covariate balance



	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.408	0.457	0.000	0.001
log_pop	10.192	9.355	0.000	0.000
gdp_pcap_cons2005	5.601	6.162	0.025	0.360
gini	0.449	0.458	0.001	0.000
etv_state_pre	3.611	1.897	0.000	0.000
etv_guerr_pre	50.457	22.301	0.000	0.000
hist_land_conflict	0.108	0.034	0.000	NA
justice_percent	0.076	0.072	0.722	0.003
rank_w_any	0.089	0.043	0.017	0.002

3 Versions of matching and weighting

3.1 Naive regression

3.2 Naive + Control regression

3.3 Matching - covariates

Throws away non-matched control units

- No replacement and one-to-one matching

- with Match(): closest X_i through normalized Euclidean distance
- BiasAdjust = TRUE: bias correction to adjust for multiple continuous variables
- other things to consider: how many matches, which algorithm (eg Coarse, pre-processing), whether to drop treatment units to remove bias, calipers to remove units with no common support

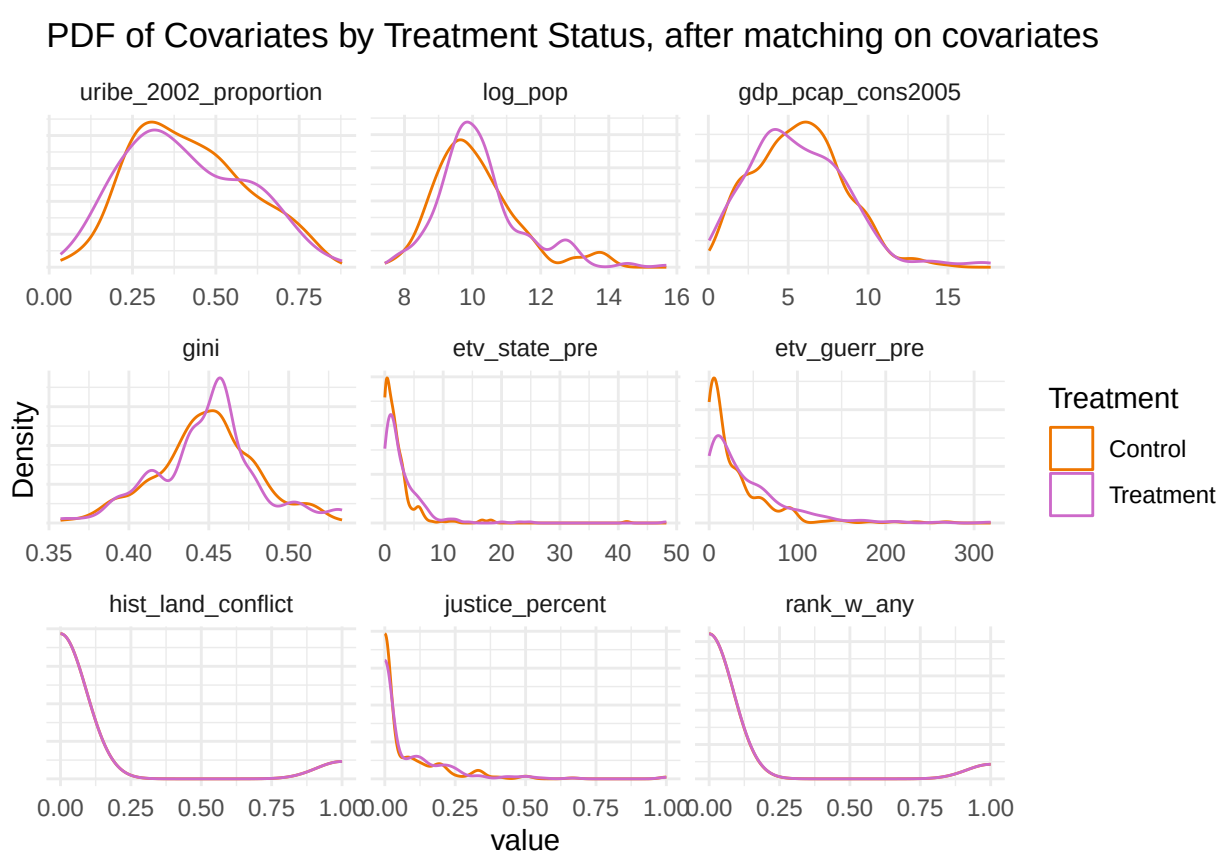
First, creating a matched dataset:

```
##
##   0   1
## 253 253
```

Should we run it multiple times? The results don't change much

Then, checking the balance after the matching

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.409	0.422	0.014	0.345
log_pop	10.204	10.075	0.000	0.159
gdp_pcap_cons2005	5.600	5.665	0.577	0.437
gini	0.449	0.449	0.513	0.468
etv_state_pre	3.073	2.150	0.000	0.000
etv_guerr_pre	41.481	28.304	0.000	0.000
hist_land_conflict	0.107	0.107	1.000	NA
justice_percent	0.075	0.068	0.070	0.494
rank_w_any	0.091	0.091	0.181	0.894



Finally, calculating ATT

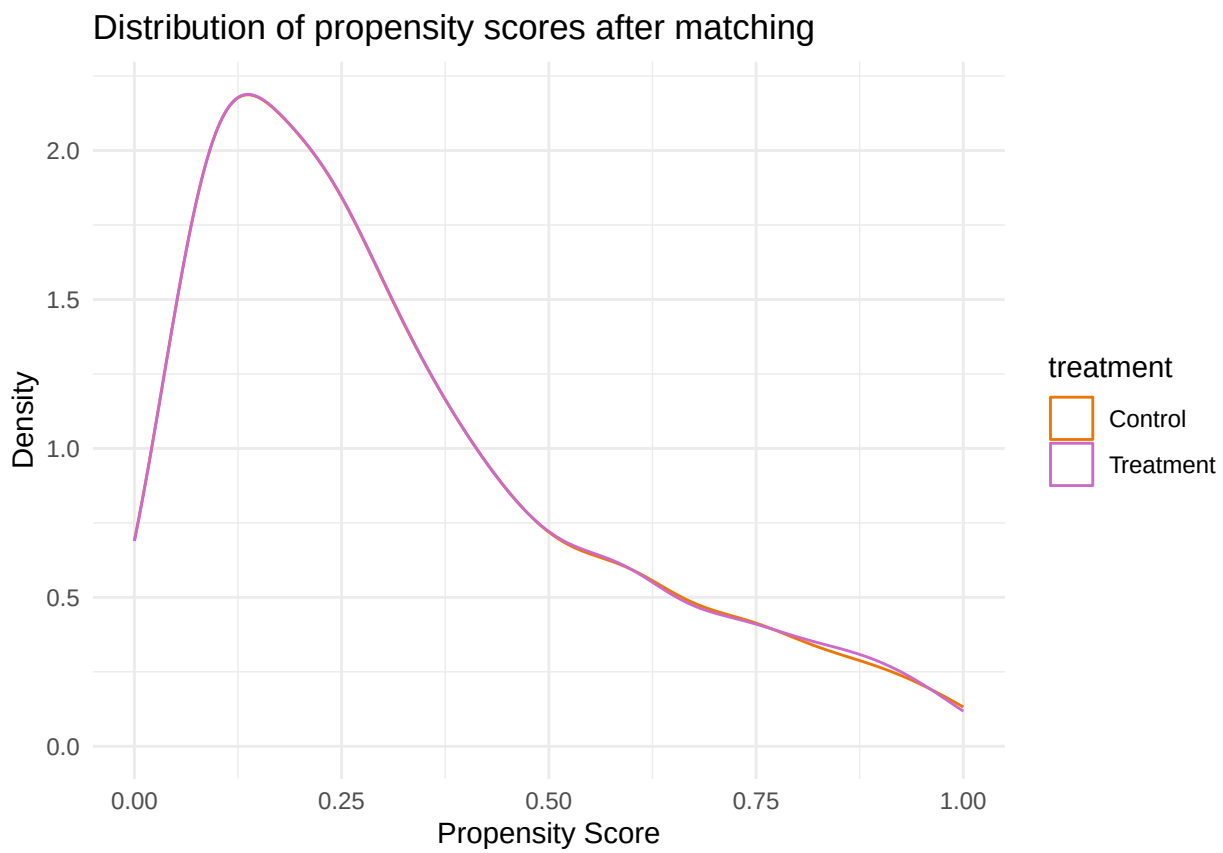
3.4 Matching - Propensity score matching

Compare units that have a comparable probability of getting treated to match:

```
##
##   0   1
## 453 453
```

Checking balance after matching:

	Treatment Mean	Control Mean	T p-value	KS p-value
uribe_2002_proportion	0.408	0.392	0.349	0.036
log_pop	10.192	9.999	0.026	0.290
gdp_pcap_cons2005	5.601	5.328	0.313	0.080
gini	0.449	0.452	0.365	0.000
etv_state_pre	3.611	3.622	0.979	0.000
etv_guerr_pre	50.457	58.058	0.179	0.000
hist_land_conflict	0.108	0.080	0.252	NA
justice_percent	0.076	0.078	0.884	0.302
rank_w_any	0.089	0.146	0.038	0.120

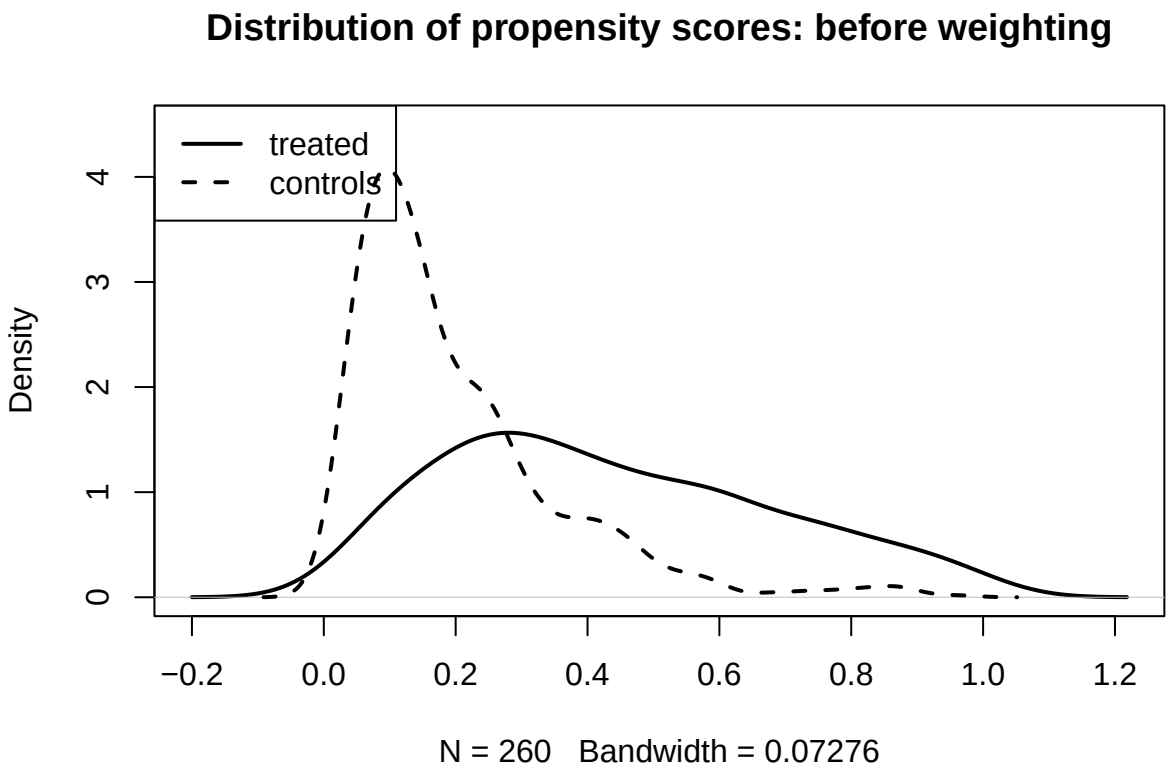


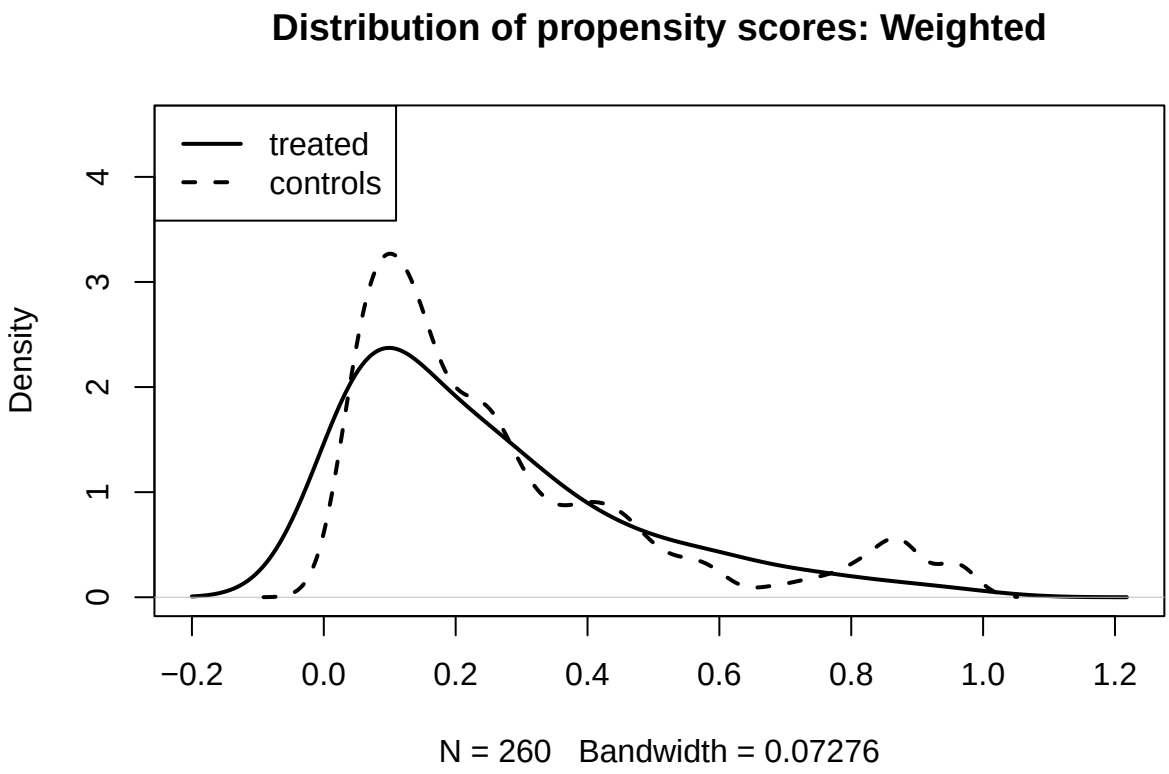
Calculating the ATT:

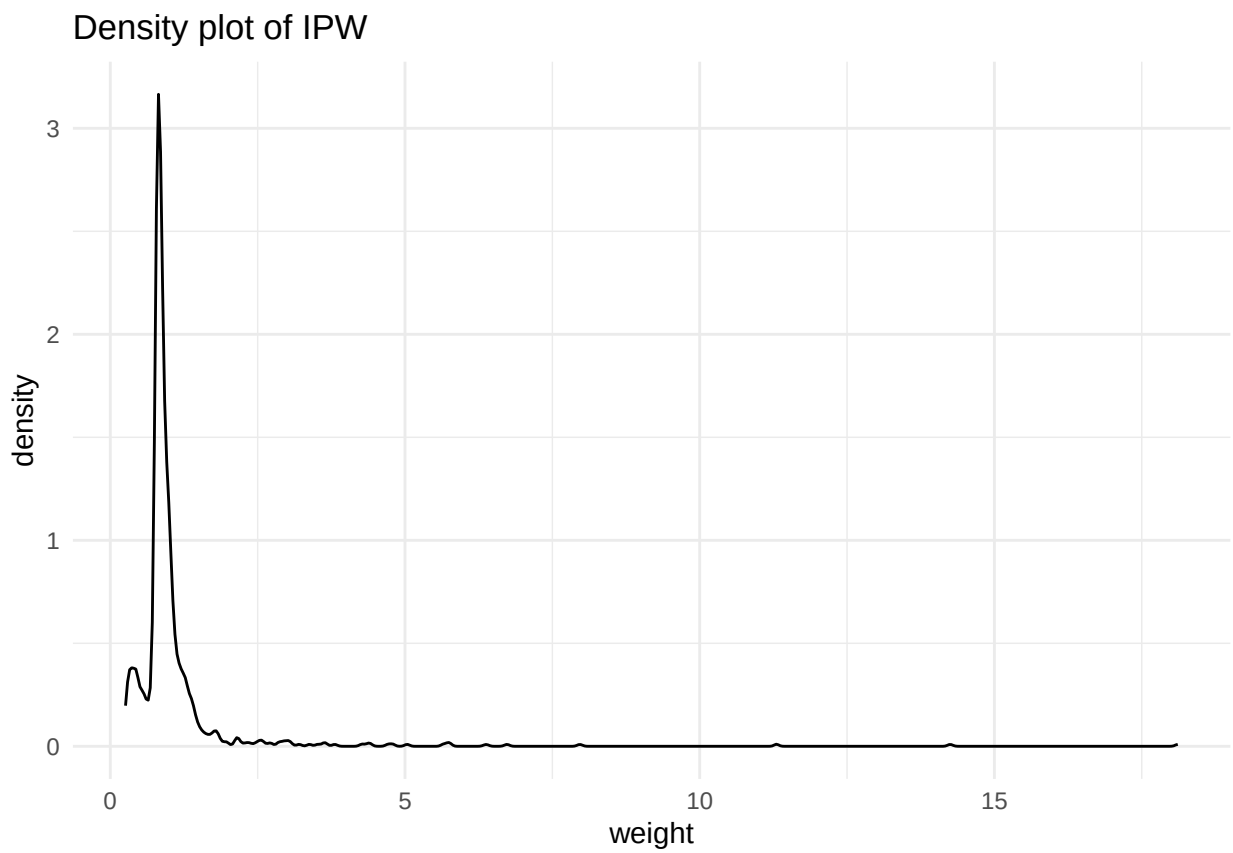
The Match() propensity scores afterwards are very close, but the data has been significantly manipulated: there are many duplicates of both control and treatment

3.5 Weighting - propensity scores

Inverse Probability-of-Treatment Weighting (IPW):


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 260 Bandwidth = 0.07276


Distribution of propensity scores: Weighted
treated
controls
Density
0
1
2
3
4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 260 Bandwidth = 0.07276

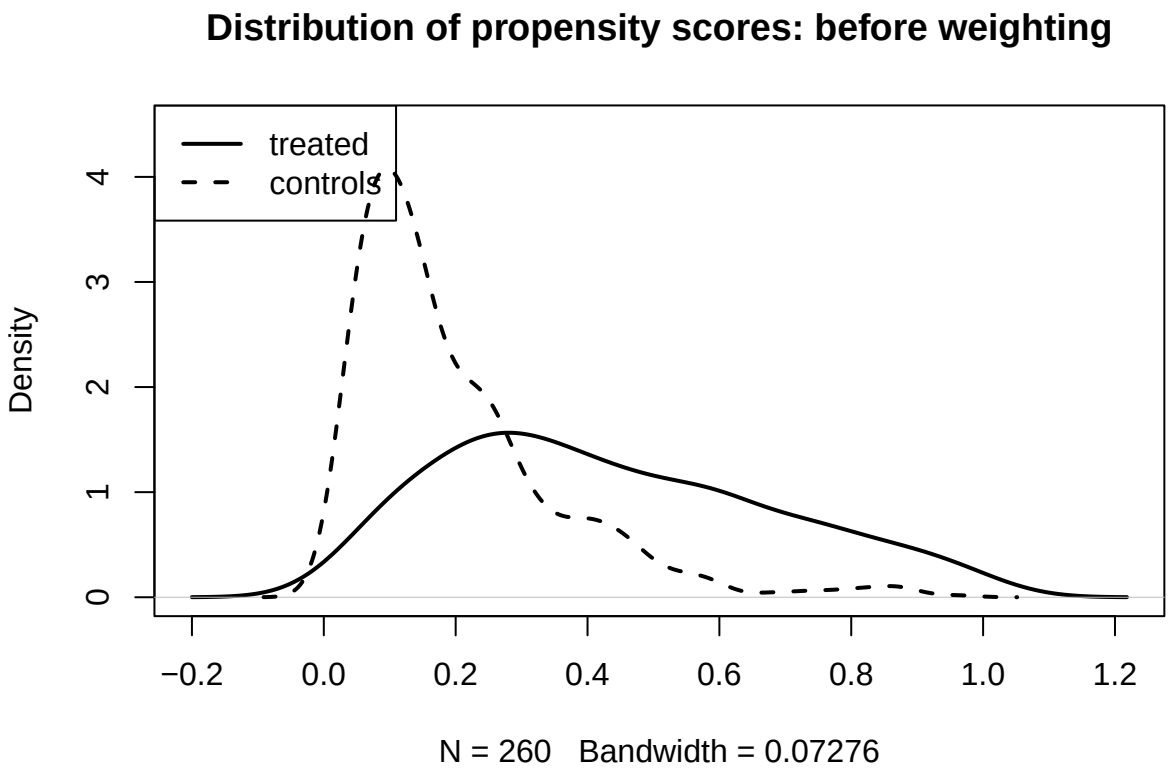


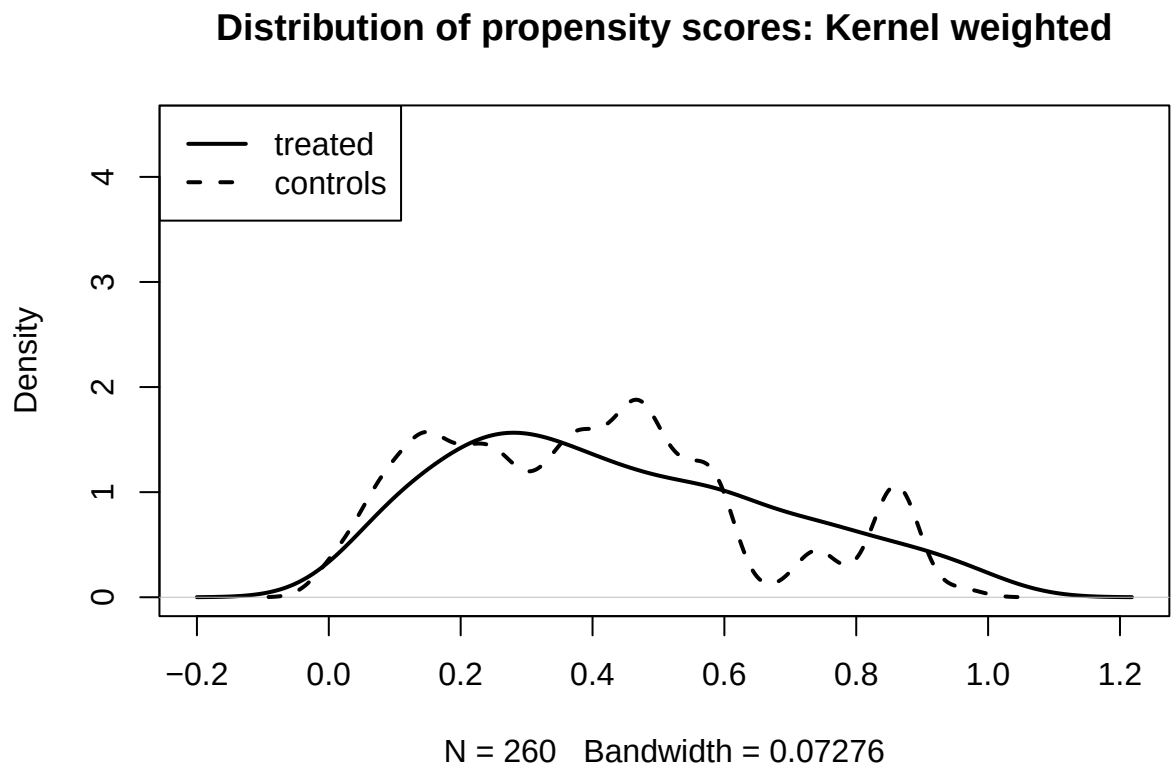
Check how extreme the weights get / how few units are doing most of the work - it means large bias

Take regressions:

3.6 Weighting - Kernel balancing weights

Propensity scores before and after kernel weighting


Distribution of propensity scores: before weighting
treated
controls
Density
0
1
2
3
4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
N = 260 Bandwidth = 0.07276

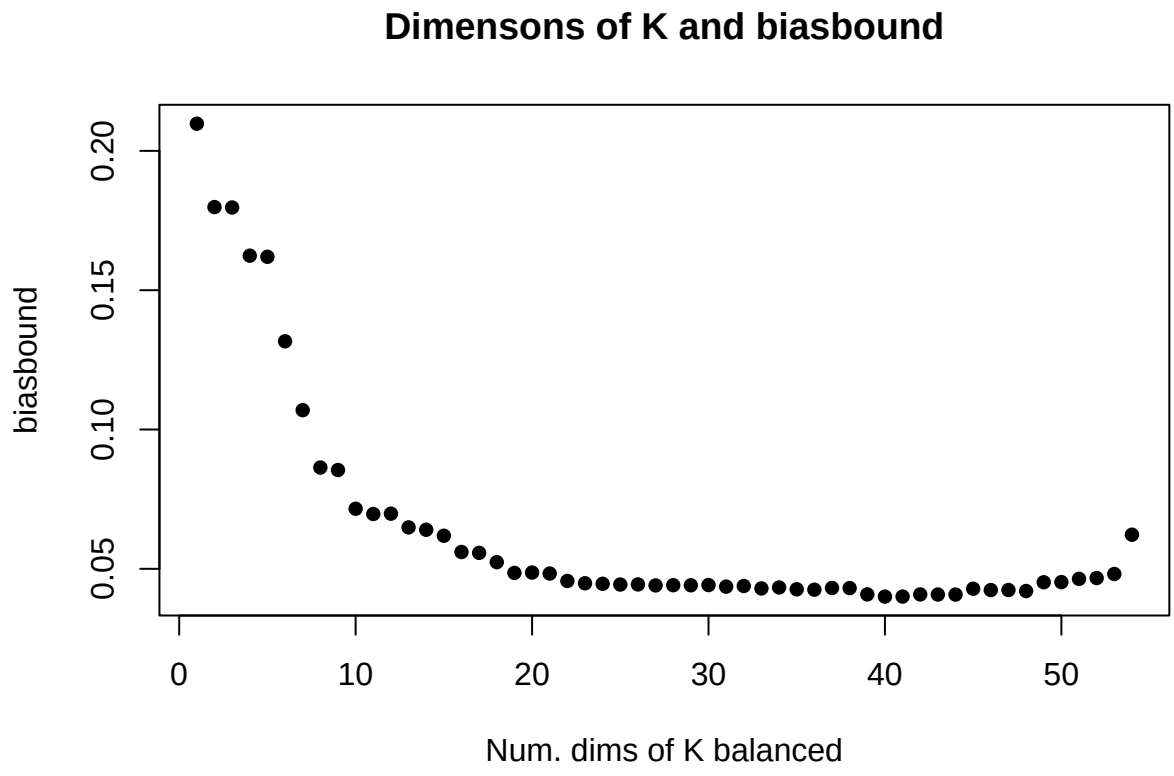


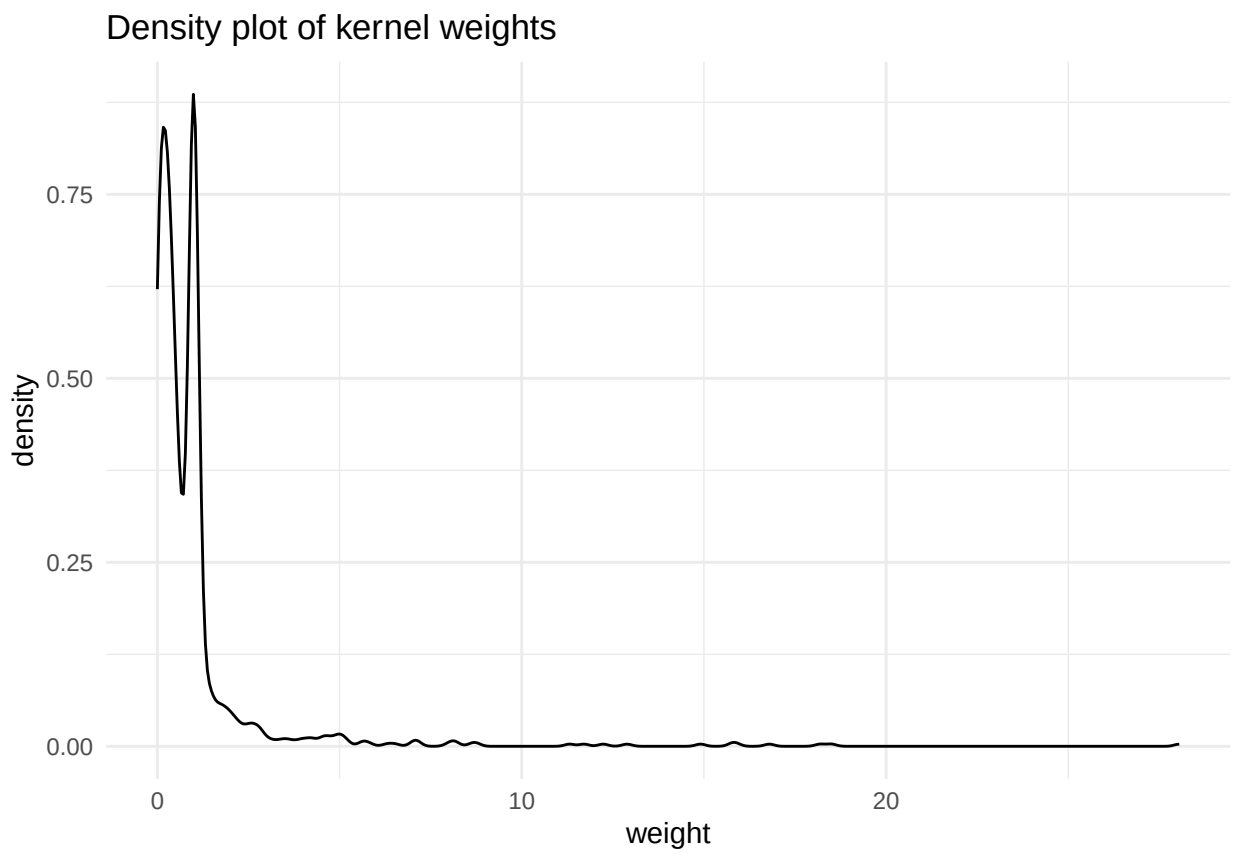
Balance tables, before and after kernel weighting

```
## Balance Measures
##                          Type  M.0.Un  M.1.Un Diff.Un
## uribe_2002_proportion Contin.  0.4565  0.4075 -0.2592
## log_pop               Contin.  9.3550 10.1916  0.7733
## gdp_pcap_cons2005     Contin.  6.1615  5.6012 -0.1472
## gini                  Contin.  0.4581  0.4495 -0.2380
## etv_state_pre         Contin.  1.8967  3.6106  0.3448
## etv_guerr_pre         Contin. 22.3013 50.4571  0.4683
## hist_land_conflict     Binary  0.0335  0.1077  0.0742
## justice_percent       Contin.  0.0722  0.0760  0.0247
## rank_w_any            Contin.  0.0429  0.0889  0.1862
##
## Sample sizes
##     Control Treated
## All     746     260
```

```
## Balance Measures
##                          Type M.0.Adj M.1.Adj Diff.Adj
## uribe_2002_proportion Contin.  0.4136  0.4075  -0.0334
## log_pop               Contin. 10.1019 10.1916   0.0726
## gdp_pcap_cons2005     Contin.  6.2527  5.6012  -0.2131
## gini                  Contin.  0.4502  0.4495  -0.0206
## etv_state_pre         Contin.  3.1151  3.6106   0.0834
## etv_guerr_pre         Contin. 40.9580 50.4571   0.1240
## hist_land_conflict     Binary  0.1054  0.1077   0.0023
```

```
## justice_percent       Contin.  0.0806  0.0760  -0.0320
## rank_w_any            Contin.  0.0861  0.0889   0.0099
##
## Effective sample sizes
##            Control Treated
## Unadjusted   746.      260
## Adjusted     119.4     260
```





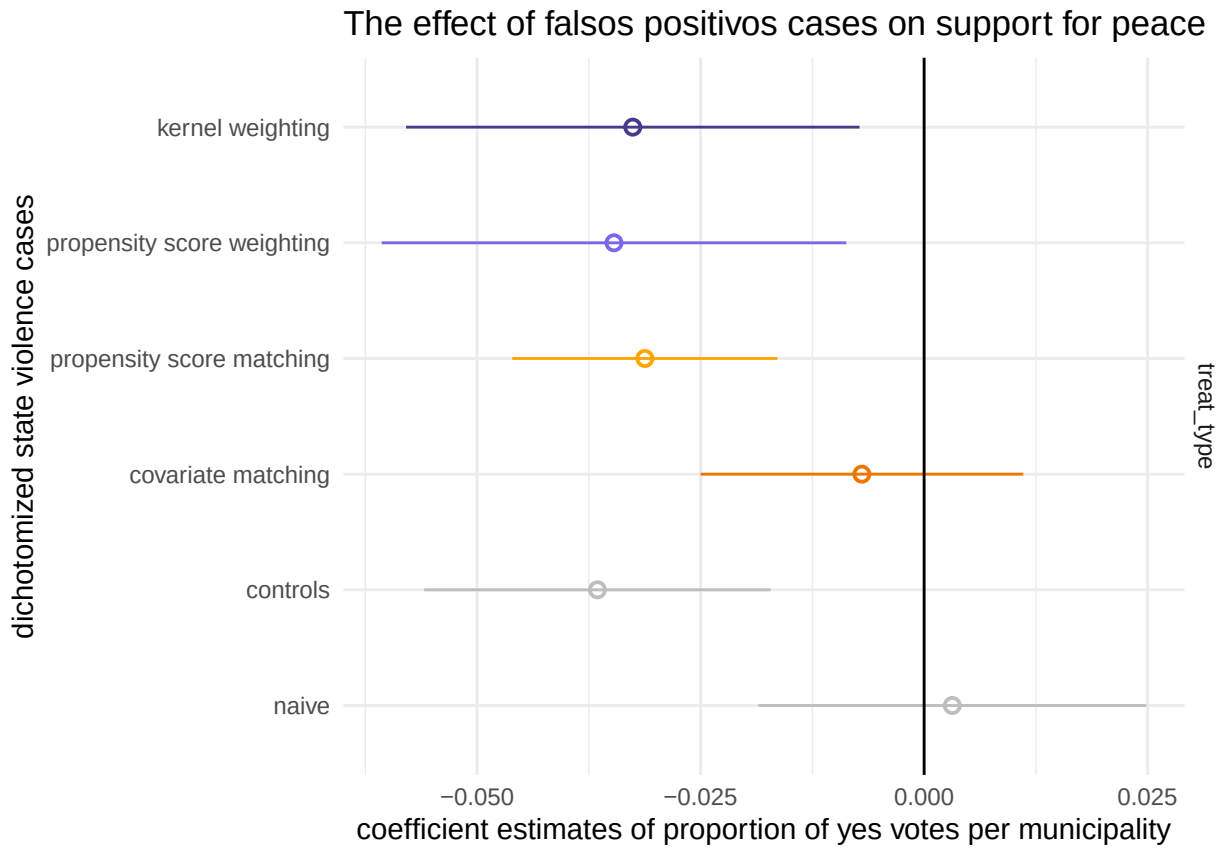
Where treatment and control are the most balanced (ebalance convergence) and bias bound is as low as possible

5. Results

Table 4: The effect of falsos positivos cases on support for peace

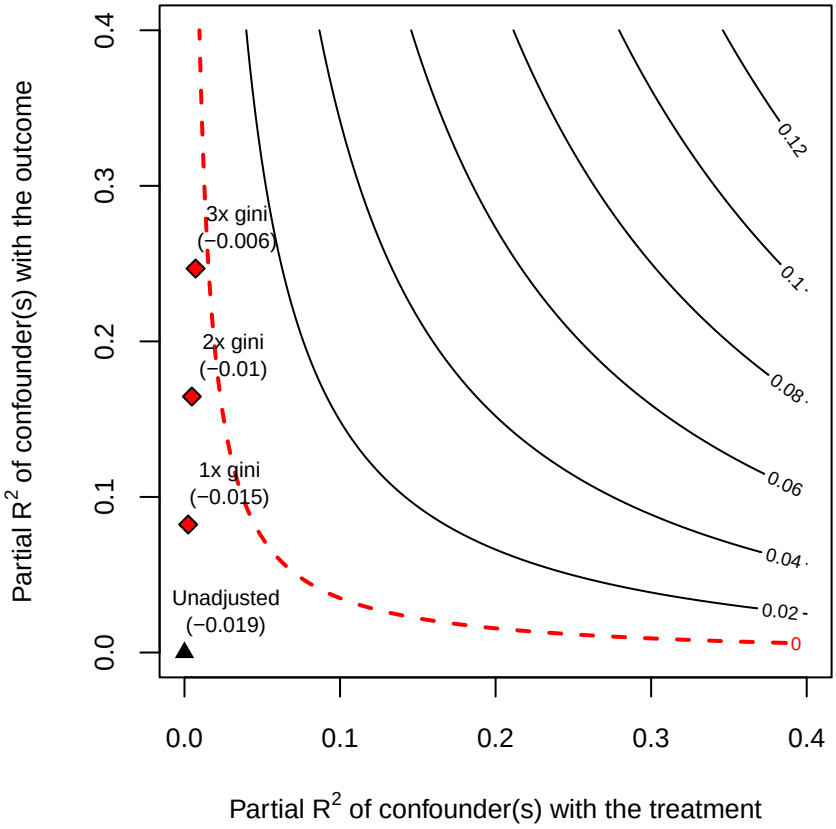
			Matching		Weighting	
	Naive	Controls	Covariates	Propensity scores		Kernel
	(1)	(2)	(3)	(4)	(5)	(6)
(Intercept)	0.493***	0.494***	0.499***	0.512***	0.501***	0.517***
	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)	(0.011)
treatment	0.003	−0.037***	−0.007	−0.031***	−0.035**	−0.033*
	(0.011)	(0.010)	(0.009)	(0.008)	(0.013)	(0.013)
uribe vote		−0.379***	−0.449***	−0.310***	−0.327***	−0.439***
		(0.027)	(0.047)	(0.032)	(0.047)	(0.053)
population		−0.015**	0.002	−0.003	−0.001	0.007
		(0.006)	(0.006)	(0.007)	(0.006)	(0.007)
gdp per capita		−0.004*	−0.010**	−0.013***	−0.007**	−0.003
		(0.002)	(0.003)	(0.002)	(0.003)	(0.003)
gini		−1.089***	−0.914***	−0.905***	−0.904***	−0.736**
		(0.152)	(0.265)	(0.202)	(0.159)	(0.273)
state violence		−0.004**	−0.002+	−0.001	−0.002	−0.003*
		(0.001)	(0.001)	(0.001)	(0.002)	(0.001)
guerilla violence		0.000*	−0.001**	0.000	0.000	0.000
		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
historical land conflict		0.007	0.005	0.045*	0.018	0.005
		(0.021)	(0.021)	(0.023)	(0.023)	(0.043)
judicial capacity		0.027	0.014	−0.014	0.012	−0.023
		(0.032)	(0.042)	(0.027)	(0.035)	(0.069)
military rank		0.067***	0.034*	0.049***	0.049*	0.048+
		(0.015)	(0.015)	(0.014)	(0.022)	(0.025)
Num.Obs.	1006	1006	506	906	1006	1006
R2	0.000	0.435	0.521	0.485	0.402	0.433
R2 Adj.	−0.001	0.425	0.502	0.474	0.391	0.422
RMSE	0.15	0.11			0.12	0.12

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001



6. Sensitivity analysis

Which covariate as a benchmark?



Sensitivity with regular lm() only; ipw/matching weights mess it up – how to add other versions?